Capstone Development Reflection Journal

Module B: AI for Leaders

Ka Tsun Hui

Weeks 2–6 · artifacts included

Week 2

Assignment/Capstone work: I picked finance, environment, and entertainment. I value derivatives at a hedge fund in Hong Kong, so finance was automatic. Environment is the EV transition, the industrial change I have watched most closely. Entertainment is music and film. I listen and I watch. I do not work in those industries.

Artifact (required): three industries, two public datasets each.

Industry	Dataset A	Dataset B
Finance	Historic options (Dubach): SPY/IWM/QQQ chains, 2008–2025	FINRA equity short interest (biweekly files)
Environment	CPCA China passenger-vehicle sales, Jan 2018–Apr 2024 (Kaggle / felixzhao)	National CO2 panel (later dropped; too far from a decision I would own)
Entertainment	Ultimate Spotify Tracks DB (Hamidani, Kaggle)	The Movies Dataset (Banik, 2017, Kaggle / TMDB)

Reflection: I almost started with the biggest files I could find. Picking industries first forced a different test: would I still care about this after the class ends? Markets, EVs, and the stuff I stream.

Week 3

Assignment/Capstone work: I pulled the SPY chain for 31 December 2024 (9,646 contracts, 23 columns) and the CPCA China auto file (38,806 model-month rows). Right away: 625 option contracts with a zero bid, duplicate model-month rows, and the string NA in body type.

Artifact (required): sample rows from each of the five datasets that ended up in Milestone One, with a one-line description. Full column lists are in the paper.

Dataset 1 : SPY options, 31 Dec 2024. One listed contract. Near-the-money sample (the file's first rows are deep in-the-money expiries).

contract_id	strike	type	last	IV
SPY250207C00594000	594	call	7.80	0.132
SPY250207P00594000	594	put	12.94	0.142
SPY250207C00595000	595	call	7.39	0.132

Dataset 2 : CPCA China passenger-vehicle sales. One model in one month.

year_month	brand	model	units	is_ev
2022-11	Tesla	Model Y	69,098	EV
2022-11	Wuling	Hongguang MINIEV	68,567	EV
2022-11	BYD	Song PLUS	64,145	EV

Dataset 3 : Spotify tracks. One track, audio features plus genre.

artists	track	popularity	danceability	genre
Gen Hoshino	Comedy	73	0.676	acoustic
Ben Woodward	Ghost - Acoustic	55	0.42	acoustic
Ingrid Michaelson;ZAYN	To Begin Again	57	0.438	acoustic

Dataset 4 : TMDB movies. One film. Grumpier Old Men shows a zero budget that is missing data, not a free film.

title	release	budget	revenue	vote
Toy Story	1995-10-30	30,000,000	373,554,033	7.7
Jumanji	1995-12-15	65,000,000	262,797,249	6.9
Grumpier Old Men	1995-12-22	0	0	6.5

Dataset 5 : FINRA short interest, 30 Sep 2026. One stock on one settlement date. File: shrt20260930.csv.

settlementDate	symbolCode	short shares	changePercent	daysToCover
2026-09-30	A	5,240,309	3.18	1.78
2026-09-30	AA	11,725,447	-2.16	2.85

2026-09-30	AAA	7,957	456.82	1.00

Reflection: The zero-bid rows looked like ordinary quotes until I asked what a zero bid is. Nobody is buying that contract at any price. Dropping 6.5% of the file felt aggressive. Using fake prices in a model is worse.

Week 4

Assignment/Capstone work: I wrote the work problems that sit on these files, then added a fifth dataset next to the options work instead of another copy of it.

The professional problems. I value derivatives at a hedge fund in Hong Kong. I kept files that map onto questions I already have to explain, or that I care about as a consumer. In finance, a quoted smile and a set of Greeks have to be internally consistent, and a short-interest file has to tell crowded positioning from names that barely trade. A bad IV surface mis-marks a book. A short-interest spike that is really a split sets off a fake squeeze alarm for risk and for stock loan. In the auto market the question is which segments went electric, and whether a subsidy cliff changed that. Wrong mix is expensive inventory and battery supply. In music and film, streamers and studios spend against noisy popularity and revenue. A popularity score that is just genre balancing, or a budget coded as zero, sends a model the wrong way. Milestone Two should stay on those questions. An accuracy number by itself is not the point.

Artifact (required): a new related dataset, with a sample and a public link. I added FINRA's 30 September 2026 short-interest file (shrt20260930.csv): one date, 22,581 names. Same day-of-market idea as the SPY chain, but positioning and days-to-cover instead of quotes. No price join.

Link: https://cdn.finra.org/equity/otcmarket/biweekly/shrt20260930.csv

settlementDate	symbolCode	issueName	currentShortPositionQuantity	daysToCoverQuantity
2026-09-30	A	Agilent Technologies	5,240,309	1.78

		Inc.		
2026-09-30	AA	Alcoa Corporation	11,725,447	2.85
2026-09-30	AAA	Alternative Access First Prior	7,957	1.00

Reflection: The module treats prediction as supervised segmentation. That is how I started to look at an option chain: boundaries across moneyness and expiry, the way a tree splits customers. After that I cared more about the shape of the problem than about which industry label it wore.

Week 5

Assignment/Capstone work: I locked five files: SPY options, China EV sales, Spotify, TMDB, and the 30 September 2026 FINRA short-interest file. Kaggle would not load on my machine, so Spotify and TMDB are local copies of the public Kaggle files. I made a slideshow: each dataset, a sample, why I kept it, what it can answer, and a tree vs linear-discriminant demo on Dataset 5. I tried SpaceX launches for environment. Too few rows. Out.

Artifact (required): slideshow of the five datasets. The slides are printed below. File: milestone1/dx699-week5-datasets.pptx.

Slide 1 of 9

Five datasets for Milestone One

DX699 AI for Leaders · Week 5 slideshow

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Slide 2 of 9

The five datasets

#	Dataset	Industry	Size	Unit
1	SPY options, 31 Dec 2024	Finance	9,646 x 23	One listed contract
2	China passenger-vehicle sales (CPCA)	Environment	38,806 x 10	One model-month
3	Ultimate Spotify Tracks DB	Entertainment	114,000 x 20	One track
4	The Movies Dataset (TMDB)	Entertainment	45,466 x 24	One film
5	FINRA short interest, 30 Sep 2026	Finance	22,581 x 14	One stock, one date

Three industries. Mixed numeric, categorical, text, and nested JSON. Dataset 5 is one FINRA settlement date, not a stacked panel.

Slide 3 of 9

1 SPY equity options

Value: a full listed chain with quotes, volume, open interest, IV, and Greeks in one table.
Why it is interesting to me: I value derivatives at a Hong Kong hedge fund. This is a direct empirical check of the volatility smile.
Questions: Can a model recover IV from moneyness and expiry? Do the Greeks behave as the textbooks say? What drives IV more, moneyness, expiry, or liquidity?

Sample (first three rows used in the paper)

contract_id	strike	type	last	IV
SPY250207C00594000	594	call	7.80	0.132
SPY250207P00594000	594	put	12.94	0.142
SPY250207C00595000	595	call	7.39	0.132

2 China passenger-vehicle sales

Value: official CPCA registrations through the EV transition, with brand, body, price band, and an EV flag.
Why it is interesting to me: the demand side of the largest industrial shift I have lived through.
Questions: Which attributes predicted electrification? Did the January 2023 subsidy end change price elasticity? Where does a seasonal baseline break?

Sample (first three rows used in the paper)

year_month	brand	model	units	is_ev
2022-11	Tesla	Model Y	69,098	EV
2022-11	Wuling	Hongguang MINIEV	68,567	EV
2022-11	BYD	Song PLUS	64,145	EV

3 Spotify track features

Value: 114,000 tracks with audio features and a popularity score, balanced across 114 genres.
Why it is interesting to me: I listen to this market every day and want to know which sonic attributes it rewards.
Questions: Which features predict popularity? Do tracks cluster into stable archetypes for playlists?

Sample (first three rows used in the paper)

artists	track	popularity	danceability	genre
Gen Hoshino	Comedy	73	0.676	acoustic
Ben Woodward	Ghost - Acoustic	55	0.42	acoustic
Ingrid Michaelson;ZAYN	To Begin Again	57	0.438	acoustic

4 TMDB movie metadata

Value: budget, revenue, votes, free text, and nested JSON for cast, crew, and genres.
Why it is interesting to me: as a film consumer I want to know what the market actually pays for before a studio spends.
Questions: What drives revenue? Can a pre-release hit/flop classifier flag failures?

Sample (first three rows used in the paper)

title	release	budget	revenue	vote
Toy Story	1995-10-30	30,000,000	373,554,033	7.7
Jumanji	1995-12-15	65,000,000	262,797,249	6.9
Grumpier Old Men	1995-12-22	0	0	6.5

5 FINRA equity short interest

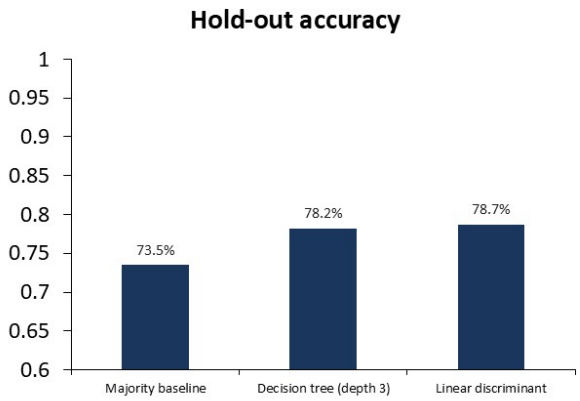
Value: 22,581 names on 30 Sep 2026 with short shares, average daily volume, and days-to-cover. One file, one date.
Why it is interesting to me: squeeze risk and borrow demand are live issues in a hedge-fund seat.
Questions: Which names are heavily shorted relative to volume? Does counting shares vs days-to-cover change who looks crowded?

Sample (first three rows used in the paper)

settlementDate	symbolCode	short shares	changePercent	daysToCover
2026-09-30	A	5,240,309	3.18	1.78
2026-09-30	AA	11,725,447	-2.16	2.85
2026-09-30	AAA	7,957	456.82	1.00

Demo: decision tree vs linear discriminant

Task: flag a name as crowded when days-to-cover > 5 on the 30 Sep 2026 FINRA file. Features: log average daily volume and market class. Zero-volume names dropped (n = 19,438). Held-out 30%. Majority class is 73.5% not-crowded.



What the tree uses
First split: log(ADV) ≤ 6.20, i.e. volume under about 500 shares a day. Low-volume OTC names are the crowded class.

Tree accuracy 78.2%. LDA 78.7%. Both beat the 73.5% majority baseline by about five points.

Week 5 point: a linear discriminant and a shallow tree find the same volume story. Neither is a trading signal. Both show the label is partly a liquidity label.

Demo in the deck: crowded means days-to-cover above 5, predicted from log volume and market class (n = 19,438 after dropping zero volume). Majority baseline 73.5%. Depth-3 tree 78.2%. Linear discriminant 78.7%. First split is log(ADV) <= 6.20, about 500 shares a day. The crowded class is low-volume OTC names. Liquidity, not a trading signal.

Reflection: I dropped more datasets than I added. The CO2 panel was the hard one. Pretty data, but I could not attach it to a decision I would own. Five files I can defend beat eight I cannot. The FINRA demo was useful for the linear-discriminant week: a shallow tree and LDA both find the same volume split, and both only beat the majority class by about five points.

Week 6

Assignment/Capstone work: I checked quality on the later files and whether regression and classification even make sense on each one. Spotify: 450 duplicate rows and 24,259 duplicate track ids. TMDB: 17 duplicates and budgets coded as zero. FINRA (one date): blank ticker on Nano Labs, days-to-cover stuck at 1.00 on 8,295 rows and at 999.99 on 4,034, and 3,143 names with zero volume. I also rebuilt the paper under the rubric labels, 12-point Times New Roman, double spacing, one page of examples.

Artifact (required): per-dataset regression / classification feasibility.

Dataset	Regression feasible?	Classification feasible?	Note
1 SPY options	Yes: quoted IV from moneyness and expiry	Yes: liquid vs illiquid from spread, volume, OI	625 zero-bid quotes must be dropped first
2 China EV sales	Weak: monthly EV units from lags is a time series; rubric discourages it	Yes: EV vs gasoline from brand, body, price	Prefer the classifier
3 Spotify	No natural numeric target I would stand behind	Yes: popularity tier or genre after de-dupe	24,259 repeated track ids are label noise
4 TMDB	Yes: log revenue from budget, genre, timing	Yes: hit vs flop from pre-release features	Zero budgets are missing, not zero
5 FINRA 30 Sep 2026	Yes: log short shares from log volume and market class (R^2 0.42)	Yes: days-to-cover > 5 from volume tier and class	Do not put short shares or days-to-cover in the classifier features

Reflection: Spotify's label noise was the puzzle I spent the most time on. Same audio features, two genre labels. You cannot classify that without first deciding what a row even is. The other headache was formatting. Double spacing cuts the word count on a page in half.